

PAPER_715: UQFF Knowledge Base 17: THz Q-Scope Signal Analysis (Set 40: Signals 31-40)

Author: Daniel T. Murphy **Date:** 2025

Class: UQFFKnowledgeBaseKB17 **CP4 Entry:** #299 **Keywords:** THz, q-scope, signals 31-40, Ug1 thread, U_bi buoyancy, flow state, UQFF KB17 **Session:** 175 | **Version:** v5.32
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Abstract

UQFF Knowledge Base 17 (KB17) presents the detailed analysis of THz q-scope Set 40 (Signals 31-40), captured at 16:46:13–16:48:10 UTC-4 on the Star-Magic q-scope apparatus. Set 40 introduces the UQFF Ug1 thread strength metric and the U_bi buoyancy adjustment as the primary physical analysis framework for THz signal bundles.

1. Instrument Configuration (Set 40)

Parameter	Value
Time/Div	200 ns
Volt/Div	500 mV
Duration	117 s (16:46:13–16:48:10)
Signals	31–40 (10 total)

2. Signal Data

Signal	Ch1 (V)	Ch2 (V)	Flow State
31	0.60	0.35	Normal
32	0.65	0.40	Normal
33	0.60	0.35	Normal
34	0.55	0.30	Chaotic
35	0.50	0.35	Inverted
36	0.60	0.40	Inverted
37	0.55	0.35	Chaotic
38	0.50	0.30	Chaotic
39	0.50	0.35	Inverted
40	0.50	0.35	Inverted

3. Ug1 Thread Strength

$$U_{g1,i} = \mu_J \omega_{THz} V_{ch1,i}^2$$

Total thread strength across Set 40:

$$\mathcal{T}_{Ug1} = \sum_{i=1}^{10} U_{g1,i} = \mu_J \omega_{THz} \sum_i V_i^2$$

4. U_bi Buoyancy Adjustment

$$U_{bi,i}(t) = \rho_{UA} \omega_{THz} V_{ch1,i} \cos\left(\frac{2\pi t}{\tau_{flow}}\right)$$

The buoyancy adjustment modulates as the Earth's core ACE/DCE pressure cycle causes flow reversals in the q-scope magnetic coupling.

5. Combined UQFF Observable

$$[U_m : SM_m / Ug1 = UQFF_g + SM_g]^{SCm} = \frac{U_{m,bundle}}{Ug1_{thread}} \cdot \frac{\rho_{SCm}}{\rho_{UA}}$$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-52} \text{ m}^{-2}$	Planck 2018	✓ Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- Q-Scope Laboratory Records Set 40, Star-Magic 2026
- UQFF Framework v5.32, Star-Magic Session 175

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